LESSON 7.13
MULTIPLYING AND DIVIDING INTEGERS

LESSON INTRODUCTION

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

LEARNING TARGETS

- I can multiply and divide integers with procedural fluency.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- How can you use the signs of the factors to determine the sign of the product?
- How can you use the signs of the dividend and divisor to determine the sign of the quotient?
- How do you multiply integers? How do you divide integers? What is similar and different?

LESSON NARRATIVE

In this lesson, students develop procedural fluency with multiplying and dividing integers. Students multiply and divide integers and determine factor pairs for given products and the dividends and divisors of given quotients.

COMPONENT SUMMARY

7.13.1 WARM-UP *(~5 MIN.)*

Analyze a temperature graph and determine a headline to describe it

7.13.2 REFRESHER *(10 MIN.)*

Write and solve multiplication and division problems based on visual models

7.13.3 COLLABORATION *(10-15 MIN.)*

Given products and quotients, determine factors, dividends, and divisors

7.13.4 GUIDED PRACTICE *(5 MIN.)*

Determine the missing factors, products, and quotients in equations

7.13.5 YOUR TURN! *(5-10 MIN.)*

Determine the products and quotients of integers

7.13.6 WRAP-UP *(~5 MIN.)*

Reflection on learning from lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Product
- Factor
- Quotient
- Dividend
- Divisor

MATERIALS

- Anchor charts

ADDITIONAL PREPARATION

- Consider preparing sentence stems for 7.13.3 Collaboration.
- Consider preparing answer choices for 7.13.4 Guided Practice.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to find the factors, given the product, and the dividends and divisors, given the quotient, if they are not confident in their math facts.

THINGS TO CONSIDER

- Consider providing sentence starters for the written responses (questions 2, 3, 4, 5, and 6) in 7.13.3 Collaboration.
- It may be helpful to provide answer choices for question 2 in 7.13.4 Guided Practice.
- In 7.13.3 Collaboration, consider initially providing one of the factors or a dividend or divisors to help students who struggle getting started.

LITERACY AND LANGUAGE ROUTINES

Turn and Talk

In 7.13.3 Collaboration, students are given products and quotients and asked to determine the factors and dividends and divisors. They are also asked to explain their work. Due to the complexity of these problems, students will benefit from pausing to check their work and answers with a partner.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

7.13.4 Guided Practice provides an opportunity to analyze samples of student work for errors. By discussing and correcting the errors, students will be less likely to make the same mistakes themselves.

DIFFERENTIATE INSTRUCTION

Utilize resources from previous lessons like manipulatives and Study Expert videos for additional scaffolding.

- Consider providing one of the factors, a dividend, or a divisor initially in 7.13.3 Collaboration.
- The 7.13.6 Wrap-Up is a reflection with open-ended questions. Consider allowing students to use an audio recorder or speech-to-text tool to share their responses.

7.13.1 WARM-UP: WHAT'S GOING ON IN THIS GRAPH?

Component Overview: *Students analyze a graph of the difference in global average temperatures from 1880 to 2020 and the middle of the 19th century. Years with averages above the average are represented with red bars. Years with averages below the average are represented with blue bars.*

TEACHER MOVES

Notice and Wonder

Ask students to share what they notice and wonder about the graph. This will give students the opportunity to formalize their ideas before completing their written responses.

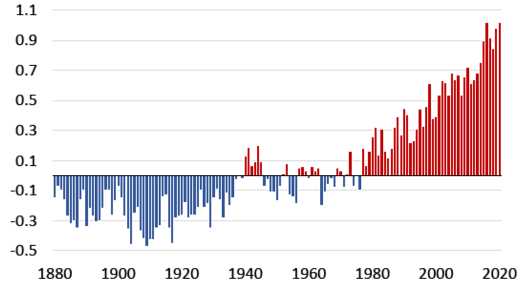
ENRICHMENT

- What do you notice about the temperatures in the 1800s and 1900s?
- What trend do you notice in the temperatures over time?

7.13.1 Warm-Up: What's Going on in This Graph?

The graph shows the average global temperatures between 1880 and 2020 compared to the middle of the 20th century. All temperatures are in degrees Celsius (℃).

Difference in Global Average Temperature



1. What are you curious about after examining the graph?
   Answers will vary.

2. Imagine this graph is included in a newspaper article. Write a catchy headline that captures what's going on in the graph.
   Answers will vary.

7.13.2 Refresher: Multiplying and Dividing Integers

1. Work with your partner to complete the tables. The last row of each table is missing both the rule and an example equation.

Answers will vary.

Visual Model of Multiplication	Sign of Product	Example Equation
+ × −	−	$3 \times (-14) = -42$
+ × +	+	$9 \times 5 = 45$
− × +	−	$-14 \cdot 4 = -56$
− × −	+	$(-7) \cdot (-8) = 56$

Visual Model of Division	Sign of Quotient	Example Equation
$\frac{-}{-}$	+	$\frac{-144}{-3} = 48$
+ ÷ −	−	$22 \div (-2) = -11$
$\frac{+}{+}$	+	$\frac{24}{4} = 6$
− ÷ +	−	$-28 \div 4 = -7$

7.13.2 REFRESHER: MULTIPLYING AND DIVIDING INTEGERS

Component Overview: *Given visual models of the signs of the factors, products, dividends, divisors, and quotients, students work with a partner to create an example equation that could be represented by the visual model. This component is meant to provide students an opportunity to review the rules they have developed for multiplying and dividing integers.*

ENRICHMENT

- How can you use the signs of the factors to predict the sign of the product? How can you use the signs of the dividend and divisor to predict the sign of the quotient?
- How can you use this to check the reasonableness of your answers when multiplying and dividing integers?

7.13.3 COLLABORATION: MULTIPLYING AND DIVIDING INTEGERS

Component Overview: *Students are given two products and two quotients. They are asked to work with a partner to determine five possible factor pairs and dividends and divisors that result in each product or quotient. Next, students explain the strategies they used.*

DIFFERENTIATE INSTRUCTION

It may be helpful to initially provide students with one of the factors and ask them to find the other missing factor.

ENRICHMENT

- How did you determine the factor pairs?
- What do you know about the signs of the factor pairs if the product is negative?
- What do you know about the signs of the factor pairs if the product is positive?

TEACHER MOVES

Turn and Talk

Have students discuss their responses to questions 2 and 3. These discussions will help students gain confidence and understanding with multiplying integers.

DIFFERENTIATE INSTRUCTION

It may be helpful to initially provide either a missing dividend or a missing divisor for students who struggle getting started. Allowing students to use a multiplication table may also be helpful.

ENRICHMENT

How are the dividends and divisors for the quotient ofthe same or different than the dividends and divisors for the quotient of ?

7.13.3 Collaboration: Multiplying and Dividing Integers

1. **The table shows two different products. Work with your partner to find five different factor pairs that, when multiplied, will result in the product. The factors cannot include 1 or -1.**
   Answers will vary.

Product	Factor	Factor
(-90)	-2	45
	3	-30
	-5	18
	-6	15
	10	-9
42	-2	-21
	3	14
	-6	-7
	21	2
	7	6

2. **Explain a strategy or an approach that each partner suggested.**
   Answers will vary. Paying attention to the sign of the product and following the rules:
   The sign of the product of integers with opposite signs is negative.
   The sign of the product of integers with the same sign is positive.

3. **Explain which of the products, (-90) or 42, was the easiest to find factor pairs for and why.**
   Answers will vary. I think 90 was easier because I knew more factors of 90 and then just had to make one factor negative and one factor positive in each pair.

4. **The table shows two different quotients. Work with your partner to find four different pairs of dividends and divisors that, when divided, result in the quotient. The numbers 1 or -1 cannot be used.**
   Answers will vary.

dividend ÷ divisor = quotient

Dividend	Divisor	Quotient
-16	2	(-8)
32	-4	
-48	6	
40	-5	
-26	-2	13
143	11	
-65	-5	
-52	-4	

5. **Explain a strategy or an approach that each partner suggested.**
   Answers will vary. Paying attention to the sign of the quotient and following the rules:
   The sign of the quotient of integers with opposite signs is negative.
   The sign of the quotient of integers with the same sign is positive.

6. **Explain which of the quotients, (-8) or 13, was the easiest to find dividends and divisors for and why.**
   Answers will vary. I think -8 was easier because it was easier for me to think of multiples of -8 to get the dividend, and then I just had to make the dividend and divisor such that one was negative and one was positive.

7.13.4 Guided Practice: Finding Missing Factors, Products, or Quotients

1. Fill in the missing values to make each equation true.

   a. $(-7) \cdot 2 = -14$

   b. $(-35) \div (-5) = 7$

   c. $\frac{-40}{8} = -5$

   d. $(-7) \times (-9) = 63$

2. Consider the equation shown. Phillip said -12 is the missing factor while Alicia claimed it is 12.

$$(-1) \cdot (-2) \cdot \underline{\ ?\ } = 24$$

Explain which student is correct using rules for multiplying integers.

Alicia is correct. $(-1) \cdot (-2) = 2$ because of the rule,

$(+) \times (+) = (+)$

Since the product 24 is positive, and applying the same rule, the missing integer is 12.

7.13.4 GUIDED PRACTICE: FINDING MISSING FACTORS, PRODUCTS, OR QUOTIENTS

Component Overview: *Students find the missing values in integer multiplication and division equations. Students apply rules of integer multiplication and division to analyze a given equation and determine which sample of student work is correct.*

DIFFERENTIATE INSTRUCTION

Consider providing sentence starters for students who may struggle explaining their work in writing. For example, "_____ is correct because of the rule ______. The correct product is _____, and the missing integer is ____."

TEACHER MOVES

Consider discussing errors during a whole class debrief if you observe similar errors throughout the class. If students are aware of common mistakes, they will be less likely to make them in their work in the future. If most students are not making errors, then have them move on to the 7.13.5 Your Turn! independently.

7.13.5 YOUR TURN!

Component Overview: *Students determine the products and quotients of integer expressions. These problems will help students build procedural fluency.*

DIFFERENTIATE INSTRUCTION

Consider your classroom when planning for the way in which you want your students engaging with these questions, if needed.

- Allow students to choose a certain number of questions to answer to increase agency and provide choice for entry points.
- Chunk all 16 questions into four groups of four by difficulty, and students must choose two of the four questions from each group.

7.13.5 Your Turn!

1. Find each product or quotient.

-11×9 -99	$-4 \cdot (-12)$ 48	$-100 \div 25$ -4	$\frac{-60}{-5}$ 12
$96 \div (-6)$ -16	$(-5) \times (-13)$ 65	$\frac{84}{-12}$ -7	$(-7)(-7)$ 49
$110 \div (-11)$ -10	$(-6)(-24)$ 144	$9 \times (-15)$ -135	-18×96 $-1{,}728$
$26 \div (-13)$ -2	$150 \div (-3)$ -50	$-39 \times (-4)$ 156	$\frac{-88}{-4}$ 22

7.13.6 Wrap-Up: Reflection

1. Complete the learning period based on today's lesson.
Answers will vary.

One question I would like to have answered …	
One mistake I learned from today …	One thing I am not sure about … YET!
One thing I am confident I know is …	

7.13.6 WRAP-UP: REFLECTION

Component Overview: *Students determine the products and quotients of integer expressions. These problems will help students build procedural fluency.*

DIFFERENTIATE INSTRUCTION

Consider allowing students to use an audio recorder or speech-to-text tool to share their responses.

